



CIS 635 Knowledge Discovery & Data Mining

Features encoding!



Outline

- Label Encoding
- One-Hot Encoding



Data/Feature Encoding

Age (Yr)	Height (M)	Weight (Kg)
10	1.3	45
30	1.7	67
65	1.5	57
...	...	...

Given a Data Table



Data/Feature Encoding

Age (Yr)	Height (M)	Weight (Kg)
10	1.3	45
30	1.7	67
65	1.5	57
...	...	...

*Numeric (R) values at
different scales*

Given a Data Table



Data/Feature Encoding

Age (Yr)	Height (M)	Weight (Kg)	Sex
10	1.3	45	F
30	1.7	67	M
65	1.5	57	F
...	...	...	...

Given a Data Table

*Categorical
Feature*

Data/Feature Encoding

Age (Yr)	Height (M)	Weight (Kg)	Sex
10	1.3	45	F
30	1.7	67	M
65	1.5	57	F
...	...	...	...

Given a Data Table

Categorical
Feature

$\{F, M\}$

$\{F=0, M=1\}$

Data/Feature Encoding

Given a Data Table

Age (Yr)	Height (M)	Weight (Kg)	Sex	Color preference
10	1.3	45	F	Red
30	1.7	67	M	Blue
65	1.5	57	F	Green
...	...	...	...	...

Categorical Feature

Data/Feature Encoding

Given a Data Table

Age (Yr)	Height (M)	Weight (Kg)	Sex	Color preference
10	1.3	45	F	Red
30	1.7	67	M	Blue
65	1.5	57	F	Green
...	...	...	...	...

Categorical
Feature

{R, G, B}

{R=0, G=1, B=2}

Data/Feature Encoding

Given a Data Table

Age (Yr)	Height (M)	Weight (Kg)	Sex	Color preference
10	1.3	45	F	Red
30	1.7	67	M	Blue
65	1.5	57	F	Green
...	...	...	...	...

$\{F, M\}$
 $\{F=0, M=1\}$

$\{R, G, B\}$
 $\{R=0, G=1, B=2\}$

Label encoder

Categorical
Feature

Data/Feature Encoding

Given a Data Table

Age (Yr)	Height (M)	Weight (Kg)	Sex	Color preference
10	1.3	45	F	Red
30	1.7	67	M	Blue
65	1.5	57	F	Green
...	...	...	...	...

$\{R, G, B\}$

$\{R=0, G=1, B=2\}$

Label encoder

Do you see any issue here?

Data/Feature Encoding

Given a Data Table

Age (Yr)	Height (M)	Weight (Kg)	Sex	Color preference
10	1.3	45	F	Red
30	1.7	67	M	Blue
65	1.5	57	F	Green
...	...	...	...	...

- We are saying, Green is closer to Red than Blue,

{R, G, B}

{R=0, G=1, B=2}

Do you see any issue here?

Label encoder

Data/Feature Encoding

Given a Data Table

Age (Yr)	Height (M)	Weight (Kg)	Sex	Color preference
10	1.3	45	F	Red
30	1.7	67	M	Blue
65	1.5	57	F	Green
...	...	...	...	...

- We are saying, Green is closer to Red than Blue,
- And enforcing Bias in the Vector Space

{R, G, B}

{R=0, G=1, B=2}

Do you see any issue here?

Label encoder

Data/Feature Encoding

Given a Data Table

Age (Yr)	Height (M)	Weight (Kg)	Sex	Color preference
10	1.3	45	F	Red
30	1.7	67	M	Blue
65	1.5	57	F	Green
...	...	...	...	...

- We are saying, Green is closer to Red than Blue,
- And enforcing Bias in the Vector Space
- Which doesn't seem ok.

{R, G, B}

{R=0, G=1, B=2}

Do you see any issue here?

Label encoder



Data/Feature Encoding

One Hot Encoding

- *How many colors do we have?*



Data/Feature Encoding

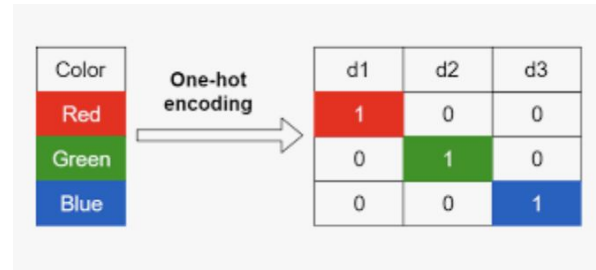
One Hot Encoding

- *How many colors do we have?*
- 3 {R, G, B}

Data/Feature Encoding

One Hot Encoding

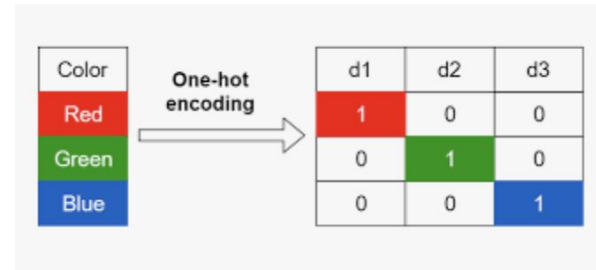
- *How many colors do we have?*
- 3 {R, G, B}
- *So, we use 3 Bits to define each color*



Data/Feature Encoding

One Hot Encoding

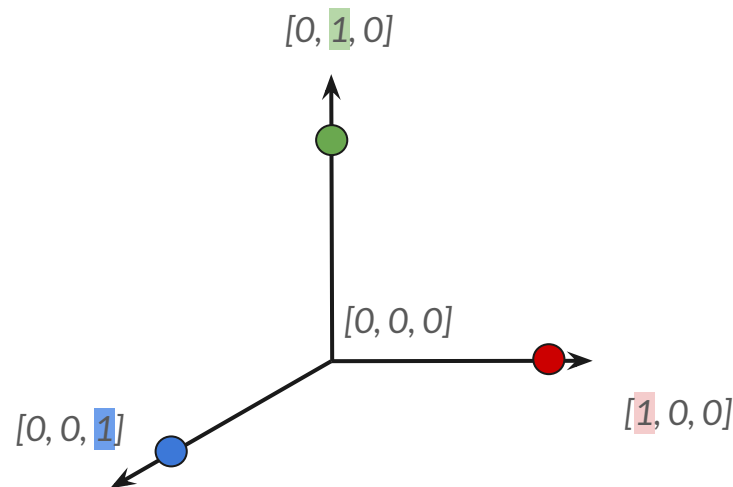
- *How many colors do we have?*
- 3 {R, G, B}
- *So, we use 3 Bits to define each color*
- *How does it solves the Vector Space Distance Problem?*



Data/Feature Encoding

One Hot Encoding

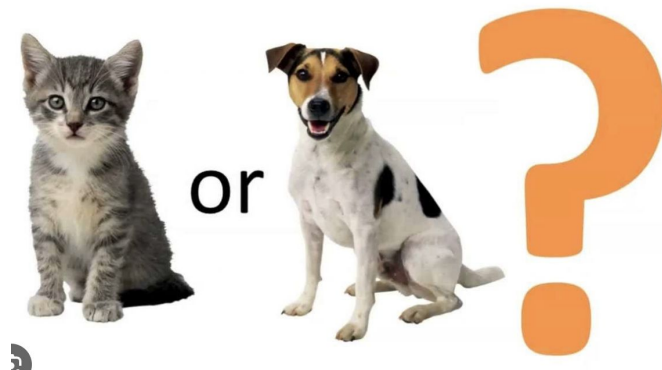
- *How many colors do we have?*
- 3 $\{R, G, B\}$
- *So, we use 3 Bits to define each color*
- *How does it solves the Vector Space Distance Problem?*
- *We have debiased the definition. Isn't it Cool!*



One hot encoding (cont.)

Classification task:

- Binary example {Cat vs Dog}
- Set size is 2
 - Cat (0, 1)
 - Dog (1, 0)
 - Or vice versa
- Same rule applies every categorical data





Notebook Presentation!

Regression task with categorical variables.



QA